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DEPARTMENT OF APPLIED MATHEMATICS
ROESSINGH RESEARCH & DEVELOPMENT

EXPLORATIVE ANALYSIS OF THREE DIMENSIONAL GAIT DATA
FOR DISTINCTIVE CLUSTERING IN WALKING BEHAVIOUR IN THE
FIGURE-OF-8 WALKING TEST

ELINE PEETERS

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1 Introduction

Roessingh Research & Development (RRD) is a research institute connected to Roessingh Center for Rehabilitation in Enschede. RRD performs scientific research on healthcare technology, among which movement analysis in rehabilitation. In order to gain insight into movements and how these are established, RRD has well-equipped facilities to register movements, external forces and muscle activity.

Currently, research projects are ongoing using clinical 3D gait assessments to gain insight into the walking dynamics of patients with impaired mobility. In the 10 meter walk test (10MWT), multiple reflective markers are placed on the patient's body, and infrared cameras register the relative 3D location of these markers. The patient is asked to walk in a 10 meter straight line, and the positional data are stored. Existing models provide extensive analysis and visualisation of this three-dimensional data. The development of metrics such as pelvic tilt or knee angle throughout the different stages of gait, give great insight into the patient's movement pattern, and the possible treatment they would need.

In comparison to daily life tasks, the 10 meter straight line is a relatively simple task. Therefore, this task can be extended to the so-called 'Figure of 8 walk test' (F8WT), as can be seen in Figure 1. In this test, patients are asked to walk around two pilons, 1.5 meters apart, each in a different direction. In the motion of turning in both directions, the patient's ability to stay in balance and symmetric becomes clear.

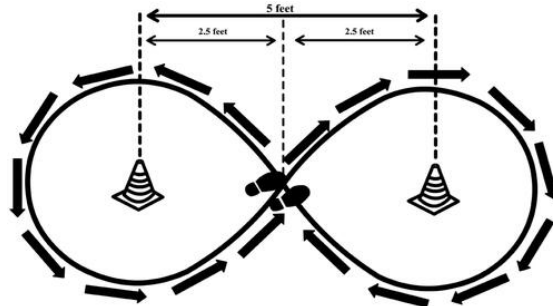


Figure 1: Set up of the Figure-of-8 Walking Test

In the standard F8WT, only the total elapsed time and the total number of steps is tracked, to give an indication of the patient's walking ability. However, extensive analysis of the F8WT may provide new insights in the movement analysis of the patient. Therefore, RRD has conducted experiments of the F8WT in their optical 3D gait analysis environment. 30 healthy subjects and 10 stroke-affected patients of different age categories are asked to perform the F8WT three times. A total of 42 markers are placed on the patient's torso, hip, knee, and foot. The relative locations of these markers are stored with a frequency of 100 measurements per second. The result is a very elaborate data set with a wide range of parameters, which can be analysed with an explorative data analysis.

2 Research question

Based on the large data set that is newly obtained, the goal is to find relationships in the kinematic and spatio-temporal data of the patients. The main objective is to find clusters of patients that show similar behaviour on different aspects of gait. This could give a better insight into what gait aspects affect balance and symmetry the most, and could clarify the walking dynamics of a curved trajectory in general. However, it is not certain that the desired relationships exist, so this study is focused on exploration of the data set. The following research question and sub-questions have been posed.

“Can relations be found in gait deviation from the Figure-of-8 walking test, and how do these relations influence dynamic balance and symmetry?”

- What gait features are introduced in the F8WT, that do not show in the 10MWT, and how can they be quantified?
- How can these gait features be clustered, to obtain significantly distinct patient groups?
- Do relationships show between clusters of different gait features?
- How can the related clusters be visualized, and what is their bio-mechanic interpretation?

3 Importance of the research

Compared to straight walking tests, the F8WT is a more accurate representation of daily life activities and the complex motor skills they require. It is a valuable combination of straight and curved path walking, and turning gait in both left and right direction. The 3D measurements give a very detailed log of the different gait phases, kinematics, and kinetics of the human body. This can give new insights into gait abnormalities that would not show in the straight path walking test. With an accurate detection of the gait abnormalities, a more accurate rehabilitation can be set up. This should help the patient in their rehabilitation process.

4 Related work

Gait kinematics in straight path walking have been extensively analysed in many researches, as well as its comparison to curved path walking. Courtine and Schieppati compared body trajectories and gait features of healthy patients in straight and curved line walk [2], [3]. They obtained data using an automatic TV-image processor. The results show that in curved segments, patients shorten the stride length of their inner leg, while the stride length of the outer leg remains the same. The outer leg has higher swing velocity and shorter stance time than the inner leg. The trunk shifts slightly inward, to create a centripetal force to enable the turn, and the feet turn inwards as well.

The comparison between the 10MWT and F8WT has also been analysed before, even for stroke patients specifically. Using inertial sensors, several spatiotemporal, stability, smoothness, and symmetry-related gait quality metrics can be determined. They show that both healthy patients and patients suffering mildly-severe stroke effects were challenged by the F8WT, where the 10MWT did not show any impairments [1]. Patients with stroke showed a lower walking speed, increased stride duration, and effected stability in the pelvis [7]. From this, it can be concluded that the F8WT gives additional insights in gait abnormalities, that the 10MWT does not show.

Most 3D clinical gait analyses with marker tracking only track a straight path, a single corner, or a single pivot turn [4], [5]. The obtained data is used to visualise the trajectory, but no researches have executed 3D measurements on the F8WT with the goal of finding relationships between the gait variables.

5 Methods

5.1 Data acquisition and normalisation

The F8WT has been executed and measured for 30 patients of different age groups. All patients do the F8WT three times, without any practise beforehand. Therefore, the first trial of each patient could be inconsistent with respect to the other two trials. If the run time of the first trial is very different from the other two, it is not included.

The walking trajectory of the Figure of 8 is not set in stone. The only instruction given to the patient is to walk around the two pions, and they can choose themselves how they execute that task. As a result, there will be many different walking trajectories in the data set. To compare and analyse the data between the patients, the trajectories need to be normalized. All variables with respect to location will be expressed as location with respect to the walking direction, which is the velocity vector of the center of mass (COM) of the body.

The obtained data goes through a biomechanical model, which returns the basic (angular) kinematics of the data points. However, these are missing for the torso, and the COM also still needs to be computed. These are added to the dataset by simple biomechanical calculations.

The F8WT consists of a single 8-shape, starting and ending in standstill (Figure 1). Therefore, the acceleration and deceleration in the first and last straight steps are different. Although this could contain additional information, analysis of these steps are beyond the scope of this research. They are cut off, such that the remaining trajectory consists of one turn left, one straight path, and one turn right.

5.2 Feature selection

Earlier research has conducted similar experiments to find clusters in (straight line) running [6]. They provide an extensive list of all kinematic and functional variables in the sagittal (y - z -)plane (Figure 2) that were tracked. However, the F8WT has a curved trajectory in the transversal (x - z) plane, which introduces shifting of the COM.

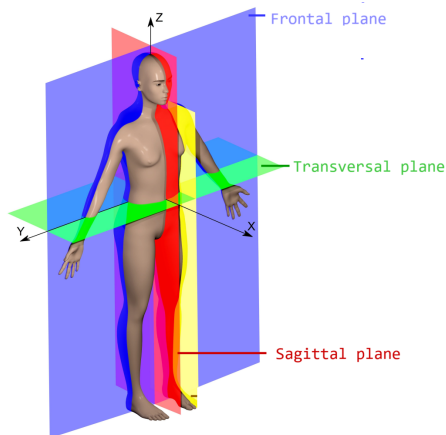


Figure 2: Anatomical planes

The smoothness and symmetry of the COM shift gives an indication of the balancing ability of the patient. To measure the balance and symmetry of the patient, measures of the foot placement with respect to the COM are added. This includes stance width, angle of rotation, and distance to the COM. In addition, the rotation, velocity, and acceleration of the different ligaments are investigated, with respect to the body direction throughout the F8WT trajectory.

In order to turn the time series data into properties of a single F8WT, the average, variance, and range throughout the test are computed for all relevant variables. These can be used to represent metrics in balance, symmetry, and smoothness [7]. Further literature study should be done to determine representative formulas and which variables are meaningful to investigate.

5.3 Clustering & Correlation

The goal of the data study is to find relationships between bio-mechanical properties in gait cycles. Cluster analysis is used to analyse the data, and discover similar clustering for different gait features. Before this cluster analysis can be applied, the gait features first need to be filtered.

5.3.1 Correlation

The selected relevant gait features can contain some double inputs. For instance, step length and step frequency are related to the step velocity. When applying cluster analysis to these gait features, they will of course show similar clustering. To be able to get an insight into new feature relationships, such similar gait features should be filtered and grouped into one.

Sets of similar gait features can easily be identified by computing the correlation coefficients for all features. A correlation heat map easily shows which gait features have high correlation with each other (Figure 3). Each set of highly correlated gait features is put together. The result is a list of unique gait features with their own properties each.

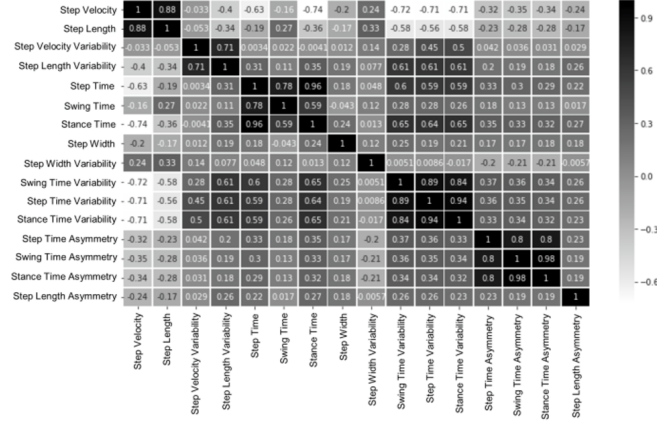


Figure 3: Example of a correlation heat map

5.3.2 Cluster analysis

After filtering the relevant gait features, patients can be grouped based on their gait pattern. Cluster analysis can be used to identify subgroups of patients with similar gait patterns. Because of the small sample size and high number of gait features, a hierarchical cluster analysis (HCA) can be used with Ward's linkage method and Euclidean distance metrics [6]. For all patient entries, the Euclidean distance is measured to all other entries. Patients with a low distance between their feature values are put in a cluster together, and close clusters are combined. This process continues until the desired number of clusters or a predetermined stopping criterion is met. An example of HCA can be seen in Figure 4A.

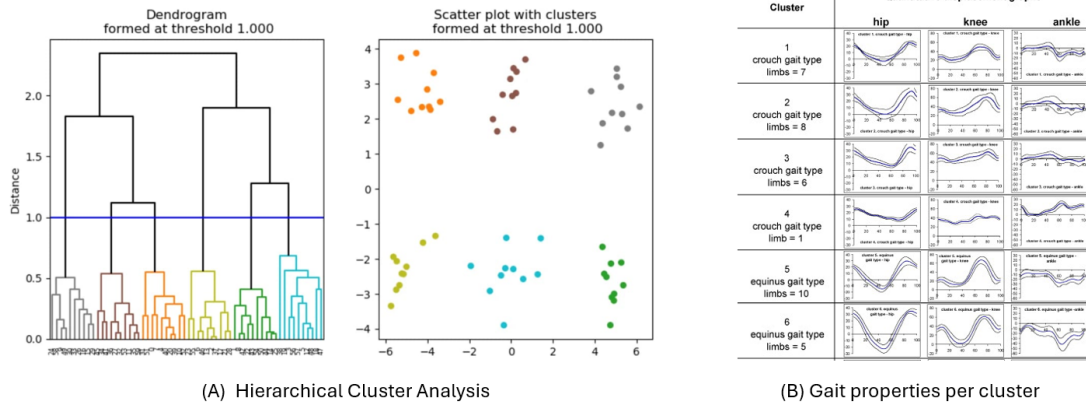


Figure 4: Example of cluster analysis and average gait comparison

Once the clusters have been computed, their properties can be analysed. For each cluster, the gate properties with significant differences with other clusters can be selected, and their (average) gait cycles can be plotted (Figure 4B). From these plots, the differences in gait cycle hopefully show some interesting bio-mechanic differences. The result is a set of unique bio-mechanic properties that represent the cluster.

A functional principal component analysis (FPCA) can be applied to give a better insight into the gait cycle. The FPCA decomposes the gait cycles of a cluster into smooth time-dependent functions, and it extracts the main functional components. An example can be seen in Figure 5. For each cluster, different principal functions should show, which should give information about the bio-mechanic differences in their gait cycles.

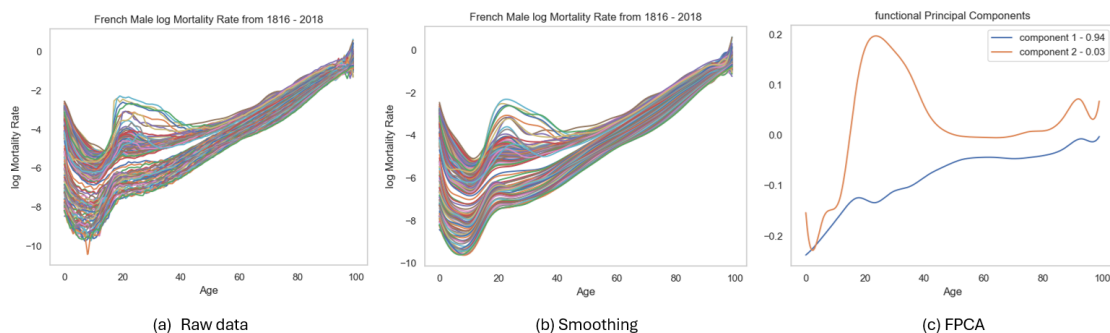


Figure 5: Example of functional principal component analysis

6 Time plan

The following time schedule gives a rough estimation of the topics per week.

	Week													
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Read papers and get acquainted with gait analysis														
Write research plan														
Extend dataset with Torso & COM kinematics														
Literature research into relevant gait features														
Gait cycle IC and FO detection														
Extend dataset with gait feature computation														
Compute correlation of gait features														
Apply code to all data & tweak IC/FO detection														
Compute and analyse clustermap of gait features														
Apply PCA to all gait features & mirror trials														
Apply PCA to feature clusters, find feature representatives														
Cluster subjects per feature cluster														
Analyse behaviour per subject clusters														
Finish report and systematically store findings														

References

- [1] Valeria Belluscio et al. “Does Curved Walking Sharpen the Assessment of Gait Disorders? An Instrumented Approach Based on Wearable Inertial Sensors”. In: *Sensors* 20.18 (2020). ISSN: 1424-8220. DOI: 10.3390/s20185244. URL: <https://www.mdpi.com/1424-8220/20/18/5244>.
- [2] Grégoire Courtine and Marco Schieppati. “Human walking along a curved path. I. Body trajectory, segment orientation and the effect of vision”. en. In: *Eur. J. Neurosci.* 18.1 (July 2003), pp. 177–190. DOI: 10.1046/j.1460-9568.2003.02736.x. URL: <https://onlinelibrary.wiley.com/doi/full/10.1046/j.1460-9568.2003.02736.x>.
- [3] Grégoire Courtine and Marco Schieppati. “Human walking along a curved path. II. Gait features and EMG patterns”. en. In: *Eur. J. Neurosci.* 18.1 (July 2003), pp. 191–205. DOI: 10.1046/j.1460-9568.2003.02737.x. URL: <https://onlinelibrary.wiley.com/doi/full/10.1046/j.1460-9568.2003.02737.x>.
- [4] Philippe C. Dixon et al. “Spatio-temporal parameters and lower-limb kinematics of turning gait in typically developing children”. In: *Gait Posture* 38.4 (2013), pp. 870–875. ISSN: 0966-6362. DOI: 10.1016/j.gaitpost.2013.04.010. URL: <https://www.sciencedirect.com/science/article/pii/S0966636213001999>.
- [5] Luis A. Nolasco et al. “The ins and outs of dynamic balance during 90-degree turns in people with a unilateral transtibial amputation”. In: *Journal of Biomechanics* 122 (2021), p. 110438. ISSN: 0021-9290. DOI: 10.1016/j.jbiomech.2021.110438. URL: <https://www.sciencedirect.com/science/article/pii/S0021929021002189>.
- [6] Angkoon Phinyomark et al. “Kinematic gait patterns in healthy runners: A hierarchical cluster analysis”. In: *Journal of Biomechanics* 48.14 (2015), pp. 3897–3904. ISSN: 0021-9290. DOI: <https://doi.org/10.1016/j.jbiomech.2015.09.025>. URL: <https://www.sciencedirect.com/science/article/pii/S0021929015005199>.
- [7] Marco Tramontano et al. “Dynamic Stability, Symmetry, and Smoothness of Gait in People with Neurological Health Conditions”. In: *Sensors* 24.8 (2024). ISSN: 1424-8220. DOI: 10.3390/s24082451. URL: <https://www.mdpi.com/1424-8220/24/8/2451>.